

## Discrete Structure

BCA — Semester 2 — — Answer Sheet

Subject Code:

**Note:** This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

### Group B

1. Explain how set operations can be represented using venn diagram with example. Convert the following sentences using quantifier. a. Not all good peoples are heroes. b. Every peoples in our country are loyal. c. Some people hate good people.

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

2. State ceiling function and floor function with examples. How mathematical induction can be used to prove the correctness of recursive algorithm? Illustrate with an example.

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

3. What does connectivity in graphs mean? Differentiate between permutation and combination. Solve the recurrence relation  $(a_n = 7a_{n-1} - 10a_{n-2})$  with initial conditions  $(a_0 = 2)$  and  $(a_1 = 3)$ .

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

4. Describe pre-order, postorder and inorder traversal of a tree with an example.

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

5. Explain one-to-one and onto function with example. What is identity function?

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

6. How can you represent relations using matrices? Explain with suitable example.

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

7. Prove that  $(\sqrt{2})$  is irrational number.

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

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**8. Solve the system of following congruences using Chinese Remainder theorem.**
$$\begin{cases} x \equiv 2 \pmod{3} \\ x \equiv 3 \pmod{5} \\ x \equiv 2 \pmod{7} \end{cases}$$

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**9. What is pigeonhole principle? Show that  $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$  where  $n$  and  $k$  are positive integers with  $n \geq k$ .**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**10. What is permutation? What is the next permutation in lexicographic order after 362541?**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**11. Explain incidence matrix representation of a graph with example.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**12. Define tree traversal. Explain pre-order traversal with example.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

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*These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.*